



News Image Caption Generation

Team 14

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Introduction & Motivation



Introduction:

- The main task is to generate captions for images using both visual content and textual information from news articles.

Motivation:

- Traditional captioning focuses only on visual features, missing broader context.
- Combining visual and textual inputs helps create more relevant and informative captions.
- Our approach aims to enhance news-image captioning by capturing the full context and relationships between image and article content.

Problem Statement



Objective:

Develop a model that integrates visual and textual inputs to generate context-aware captions:

Visual Features (I_f) are extracted from images using attention mechanisms.

Textual Features (A_t) are derived from news articles through embeddings and attention mechanisms.

$$Caption = f(I_f, A_t)$$

This approach aims to generate captions that align both with the image content and the article context, and generate captions based on a broader context

Literature Review



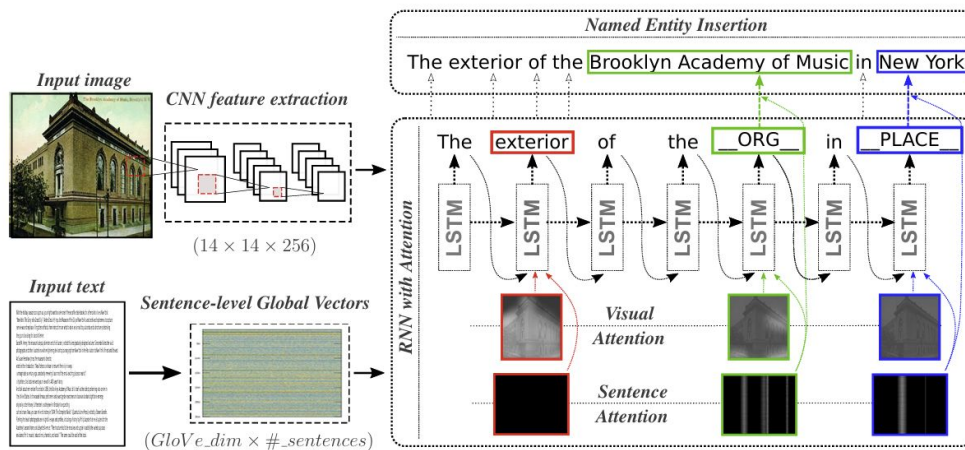
Knowledge Prompt Makes Composed Pre Trained Models Zero-Shot News Captioner

- The models approach uses context-aware knowledge prompts with pre-trained language models for generating named entities in captions.
- In this method they combine complementary pre-trained models, avoiding the need for extensive training from scratch.
- **Implementation:**
 - Flan-T5 for commonsense reasoning.
 - GPT-2 for language modeling, processing up to 512 article tokens with selected knowledge assertions.
- The model outperforms zero-shot baselines and shows competitive results without fine-tuning on news datasets.

Literature Review

Good News, Everyone!

Context driven entity aware captioning for news images



Named Entity Insertion is done using three techniques: Random Insertion, Context-based, and Attention-based Insertion (AttnIns).

For **Template Generation** the model uses softmax and LSTM to predict words, combining attended image features and article features.

Article Encoding:

Follows sentence-level encoding because to **preserve context**. Uses **GloVe embeddings** to encode text and focus on relevant sentences **through attention**.

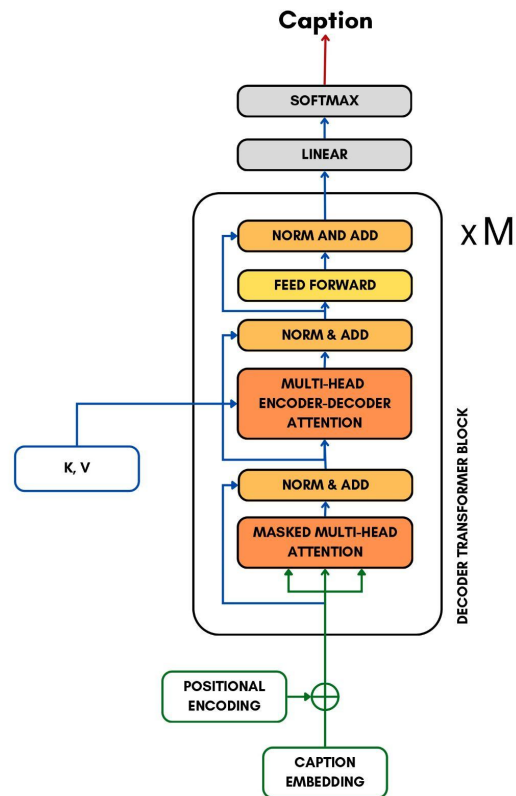
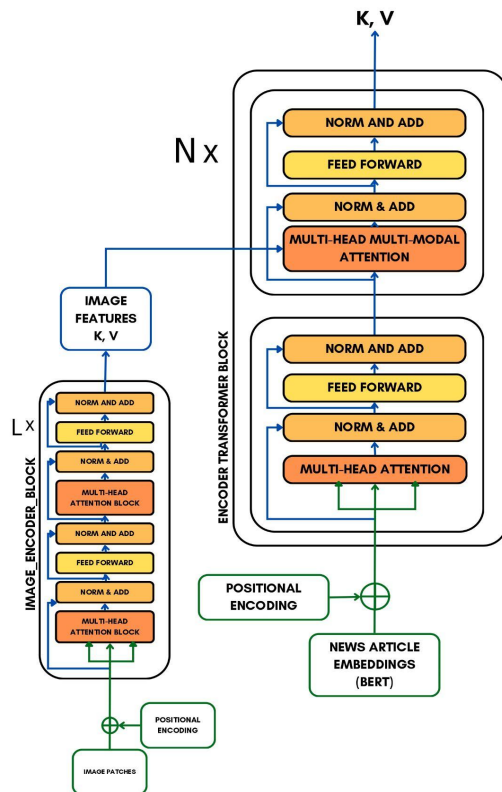
Implementation



GoodNews Dataset

- Large and diverse dataset with 466,000 images with captions and articles.
 - High-quality captions written by professional journalists of New York Times Newspaper.
 - Named entities comprise of 20% of each caption enhance contextual understanding.
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- Many instances have incomplete texts and leading sentences, missing key information.
 - Limited to articles published between 2010–2018, missing recent developments.

Implementation

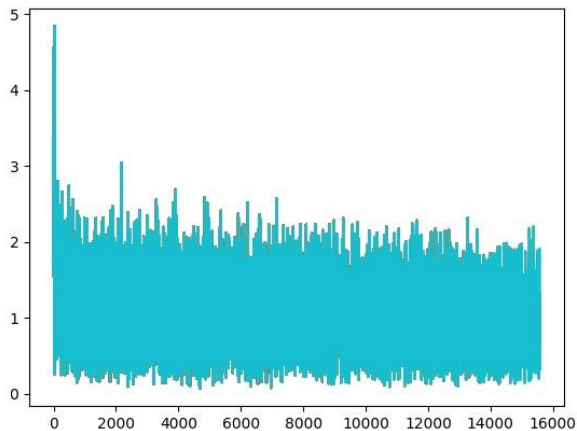


Implementation

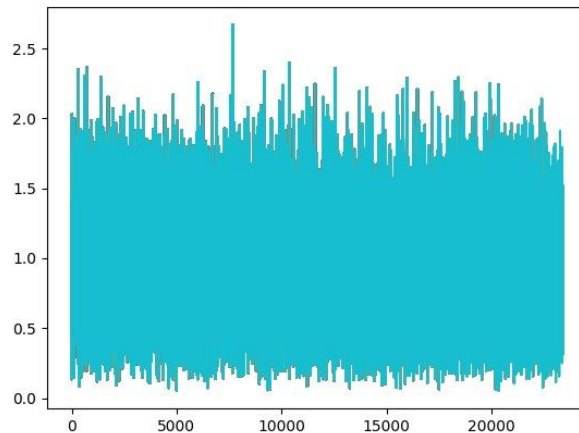


RESULTS:

Iterations vs Batch loss



25 ep * lr= 1e-4



10 ep * lr= 1e-4 + 15 ep * lr= 5x1e-5

Sample



A bit of silver lining at the onset of the Trump administration has been the groundswell of activism by hundreds of thousands of Americans taking to the streets to face off against a president, and a Republican establishment, they see as reckless, divisive and destructive.

But Republican lawmakers have been so alarmed by the size and intensity of the demonstrations -- for women's rights, immigrants and the environment -- and by protests around the country before the election, that they have introduced measures in at least 10 states to intimidate free speech by criminalizing it.

While their proponents say the bills and initiatives are needed to protect public safety and ensure civility, these efforts would crush the right of free protest at a time when key American principles and institutions are under attack.

In North Dakota, where activists have fought the construction of an oil pipeline that threatens to pollute a neighboring Indian reservation's drinking water as well as deface sacred lands, lawmakers have proposed punishing protesters who demonstrate on private property with up to 30 days in jail and a \$1,500 fine. Another state measure would make wearing a mask in a public setting a crime punishable by up to a year in prison.

Original: People protesting in North Dakota last November.

Predicted: a polling station in san francisco on wednesday. the company has been a wages that the a economy of a recent - year, from the united states of the the.

Learnings



Gained insights into challenges like integrating multimodal data, implementing complex transformer architecture.

Better understanding of time series models like RNN

Basic exposure to NLP techniques such as BERT tokenizer, GLoVe embeddings, Named Entity Recognition etc.

Improved skills in implementing pytorch

Enhanced teamwork, project management, and collaborative debugging skills

Individual Contributions



- Base Architecture code (Image only) : Devashish, Srinith
- Extending to multimodal architecture (Integrating articles): Girish, Tushita
- Paper Reviews:
 - Devashish: Goodnews
 - Girish: Knowledge paper
 - Srinith, Tushita: Multimodal transformer paper
- Implementing good news paper

References



[Paper reference](#)

[Paper reference](#)

[Image reference](#)

Literature Review



Image Caption Generation for News Articles using Multimodal Transformer

- The model takes an image and a sequence of text tokens from an article as input, and generates a sequence of tokens as the caption.
- The **Image Encoder** converts the image into a feature vector using a CNN, while the **Image-Article Encoder** computes joint representations of the image and text using an attention mechanism.
- The encoder module applies multi-head self-attention followed by a feedforward layer, and the image-attending module uses multi-head target-source attention to refine image-text interaction.
- The proposed multimodal model combines image features from ImageNet and Places365, and performs better in generating descriptive captions compared to the text-only model, which focuses on **grammatical accuracy**.

Learnings



- Implementation in PyTorch
- Exposure to Multi-Modal Learning through Transformers
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